

Question 1:

Explain CSS Grid and how it differs from Flexbox. When would you use Grid over Flexbox?

- CSS Grid is a layout system used to design two-dimensional layouts (rows and columns).
 - It allows precise control over both horizontal and vertical alignment of elements.
- **Difference from Flexbox:**
 - **Flexbox:**
 - Works in one direction (row or column).
 - Best for aligning items within a line.
 - **Grid:**
 - Works in two directions (rows *and* columns).
 - Best for designing entire page or section layouts.
- **When to Use Grid:**
 - When you need a complex, two-dimensional layout (e.g., web page structure).
 - Use Flexbox for simple, one-dimensional alignment (e.g., nav bars, buttons in a row).

Question 2:

Describe the grid-template-columns, grid-template-rows, and grid-gap properties. Provide examples of how to use them.

- **CSS Grid Properties:**
 - **grid-template-columns**
 - Defines the number and width of columns in a grid.
 - Example:
 - grid-template-columns: 100px 200px auto;
 - Creates 3 columns with widths 100px, 200px, and automatic.

→ **grid-template-rows**

- Defines the number and height of rows in a grid.
- Example:
 - grid-template-rows: 100px 150px;
- Creates 2 rows with heights 100px and 150px.

→ **grid-gap (or gap)**

- Sets the space between rows and columns.
- Example:
 - grid-gap: 10px;
- Adds 10px space between all grid items.

➤ **Example:**

```
.container {  
  display: grid;  
  grid-template-columns: 1fr 2fr;  
  grid-template-rows: 100px 200px;  
  gap: 10px;  
}
```